INCH-POUND

A-A-59429
20 APRIL 1999

COMMERCIAL ITEM DESCRIPTION

RIVETS, BLIND, NONSTRUCTURAL, RETAINED MANDREL, OPEN END, 120° COUNTERSUNK HEAD, ALUMINUM ALLOY, CARBON STEEL, NICKEL-COPPER ALLOY

This General Services Administration has authorized the use of this Commercial Item Description for all Federal agencies.

1. SCOPE: This Commercial Item Description (CID) covers the requirements for pull-stem expandable blind rivets. These rivets are intended for rapid and permanent non-structural fastening applications in which access is available to one side only.

2. SALIENT CHARACTERISTICS:

2.1 Blind rivets furnished under this CID shall conform to the materials, strengths, and dimensions as specified in Table I, II, and III and as shown in Figure 1 below.

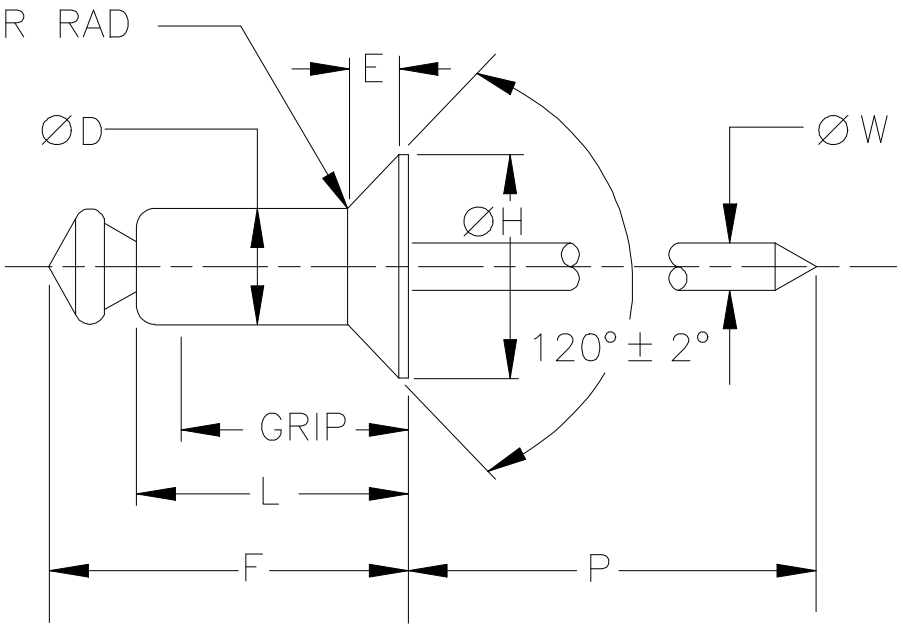

FIGURE 1. RIVETS

BENEFICIAL COMMENTS, RECOMMENDATIONS, ADDITIONS, DELETIONS, CLARIFICATIONS, ETC., AND DATA WHICH MAY IMPROVE THIS DOCUMENT SHOULD BE SENT TO: DEFENSE INDUSTRIAL SUPPLY CENTER, ATTN: DISC-AEEB, 700 ROBBINS AVENUE, PHILADELPHIA, PA 19111-5096.

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TABLE I. MATERIAL COMBINATION SYMBOLS AND STRENGTHS

SYMBOL	MATERIAL COMBINATION	RIVET DIA	MINIMUM SHEAR LOAD LBS	MINIMUM TENSION LOAD LBS
A	Body: 5056 Al Alloy (UNS A95056) Mandrel: 1006-1038 Steel (UNS G10060-G10380)	.125 .156 .188	170 260 380	220 350 500
B	Body: 5052 Al Alloy (UNS A95052) Mandrel: 2024 Al Alloy (UNS A92024)	.125	120	150
D	Body: 1006-1010 Steel (UNS G10060-G10100) Mandrel: 1030-1060 Steel (UNS G10300-G10600)	.125	260	310
E	Body: Nickel Copper Alloy (UNS N04400) Mandrel: 1030-1060 Steel (UNS G10300-G10600)	.125 .156 .188	350 550 800	450 700 1000

TABLE II. BASIC DIMENSIONS

NOM DIA	BASIC DIMENSIONS							RECOMMENDED HOLE SIZE
	ØD	∅H	E REF	R MAX	∅W NOM	F MAX	P MIN	
.094	.096 .090	.187 .161	.027	.020	.057	L+.100	1.00	.097-.100
.125	.128 .122	.233 .207	.031	.025	.076	L+.120	1.00	.129-.133
.156	.159 .153	.294 .268	.040	.030	.095	L+.140	1.06	.160-.164
.188	.191 .183	.361 .335	.050	.035	.114	L+.160	1.06	.192-.196

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TABLE III. DASH NUMBERS AND DIMENSIONS

DASH NUMBER				L MAX	RECOMMENDED GRIP RANGE
A	B	D	E		
A302		D302		.250	.079-.125
A304	B304	D304		.375	.126-.250
A401	B401		E401	.203	.040-.062
A402	B402	D402	E402	.275	.063-.125
A403	B403	D403	E403	.337	.126-.187
A404	B404	D404	E404	.400	.188-.250
A405	B405	D405	E405	.462	.251-.312
A406	B406	D406		.525	.313-.375
A502	B502	D502	E502	.290	.063-.125
A503	B503	D503	E503	.362	.120-.187
A504	B504	D504	E504	.425	.188-.250
A506	B506	D506	E506	.550	.251-.375
A508	B508	D508		.675	.376-.500
A602	B602	D602	E602	.315	.063-.125
A604	B604	D604	E604	.450	.126-.250
A606	B606	D606	E606	.575	.251-.375
A608	B608	D608	E608	.700	.376-.500
A610	B610	D610	E610	.825	.501-.625
A612				.950	.626-.750

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2.2 Materials shall be of the type and grade specified in Table I.

2.3 Surface finish.

2.3.1 Aluminum alloy rivets shall be furnished either chemically treated in accordance with MIL-C-5541, or anodically treated in accordance with MIL-A-8625 at the supplier's option

2.3.2 Carbon steel bodies shall be zinc plated in accordance with ASTM B633, Type II, FE/ZN5 or phosphate coated in accordance with Type Z, Class 2 or 4B of DOD-P-16232. Carbon steel mandrels shall be zinc plated or phosphate coated.

2.3.3. Nickel-Copper Alloy bodies shall be zinc plated in accordance with ASTM B633, Type II, FE/ZN5.

2.4 Workmanship. Rivets shall be of uniform quality and free from injurious seams and other injurious defects.

2.5 Single shear load. The shear test shall be performed in accordance with Test Method 20 of NASM1312, or equivalent and shall conform to the data specified in Table I.

2.6 Tension load. The tension test shall be performed in accordance with Test Method 8 of NASM1312, or equivalent and shall conform to the data specified in Table I.

2.7 Packaging requirements The requirements for packaging shall be in accordance with ASTM D3951.

3. QUALITY ASSURANCE PROVISIONS.

3.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and the quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

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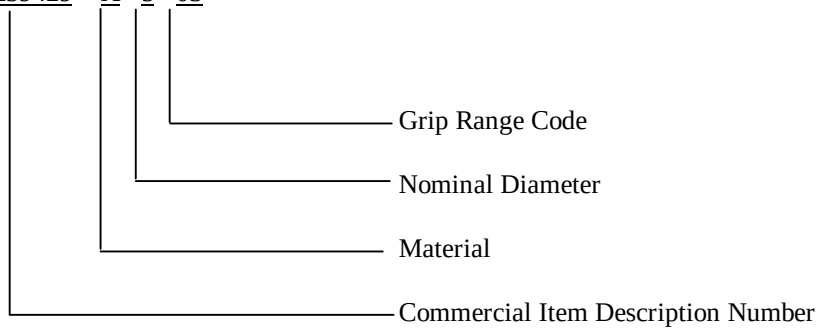
4. NOTES

4.1 All dimensions are in inches.

4.2 The part number shall consist of the basic Commercial Item Description Number followed by a dash number from Table III. The dash number consists of a material combination symbol from Table I, a nominal body diameter size code in thirty-seconds of an inch from Table II, and a maximum grip code in sixteenths of an inch from Table III.

EXAMPLE: AA59429 A 5 03



AA59429A503 Indicates - Rivet, blind, nonstructural, retained mandrel, open-end, 120° countersunk head, 5056 (UNS A95056) aluminum alloy body, C1006-C1038 (UNS G10060-G10380) steel mandrel; nominal body diameter size, .156; grip range, .120-.187.

Military Interests:

Custodians:
Army - AR
Navy - AS
Air Force - 99

Preparing Activity:
DLA-IS

Review Activities:
Army - AV, MI
Air Force - 11, 82
National Security Agency-NS

(Project No. 5320-0958)

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2. The submitter of this form must complete blocks 4, 5, 6 and 7.
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NOTE: This form may not be used to request copies of documents, or to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-59429	2. DOCUMENT DATE (YYMMDD) 990420

3. DOCUMENT TITLE:

RIVETS, BLIND, NONSTRUCTURAL, RETAINED MANDREL, OPEN END, 120^{O} COUNTERSUNK HEAD, ALUMINUM ALLOY, CARBON STEEL, NICKEL-COPPER ALLOY

4. NATURE OF CHANGE

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY DLA-IS

a. NAME Emelia Altomari	b. TELEPHONE (Include Area Code) (1) Commercial (215) 697-6827 (2) (DSN) 442-6827
c. ADDRESS (Include Zip Code) Defense Industrial Supply Center 700 Robbins Avenue Bldg. 3 (Code DISC-AESD) Philadelphia, PA 19111-5096	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041 TELEPHONE (703) 756-2340 AUTOVON 289-2340

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